Capstone Project

Predicting Diabetes From Health Related Data

By

Connor David Miller

14/08/2026

Contents

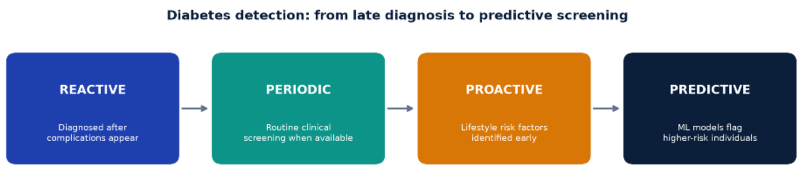
Problem Statement

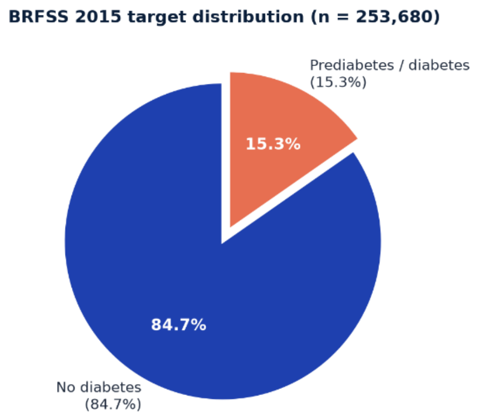
In 2015 about 415 million people worldwide were living with diabetes, and this number is expected to exceed 640 million by 2040 (Papatheodorou et al., 2018). Many people are unaware they have diabetes, which raises the risk of later health complications (Bajaj et al., 2025). Diabetes prevalence is also linked to obesity, diet, and physical inactivity (Corkey, 2012).

This project looks at whether health and lifestyle indicators from population survey data can identify people at higher risk of diabetes early enough to support targeted screening. We use data already collected at a large population scale to identify factors linked to increased diabetes risk.

The Centres for Disease Control and Prevention (CDC) Behavioural Risk Facto Surveillance System (BRFSS) 2015 dataset, only had 15.3% of respondents labelled as having prediabetes or diabetes, while 84.7% did not have diabetes (see the Pie Chart Below). Diabetes Screening is now moving from reactive diagnosis toward earlier, data-driven risk identification (see the diabetes detection below). The question then becomes whether survey data can support earlier screening detection, not clinical diagnosis.

The desired outcome of this project is to build a practical screening model with strong recall and fewer missed cases. It would not replace clinical diagnosis.





Related studies to the project show BRFSS-style survey data can support machine learning models predicting diabetes risk. Xie et al. (2019) and Majcherek et al. (2025) found BMI, age, and general health among the strongest predictors within the machine learning models. The difference in this project is we compare models and feature sets with a focus on recall and false negatives, not accuracy alone.

Industry/ domain

We are working in public health and population health, focusing on diabetes risk screening and chronic disease prevention. Rather than diagnosing diabetes clinically, we attempt to identify higher-risk people earlier using U.S. health and lifestyle survey data collected at population scale. BRFSS already collects useful indicators across U.S. adults (CDC, n.d.). The challenge is using that data to decide who should be screened first. Survey, clinical, or screening data can flag people for confirmatory testing such as HbA1c or fasting glucose (Inzucchi, 2012), then prevention or treatment.

Key concepts in this project are prevention, early detection, and screening versus diagnosis. Because missing someone at risk is more significant than an unnecessary follow-up, we prioritise recall and false negatives.

Stakeholders

The main stakeholders in this project are public health agencies, GPs and nurses, policymakers, and patients.

Public health agencies want practical use of survey data to target screening. GPs and nurses want a clear risk flag that supports triage, keeps false negatives low, and does not replace diagnosis. Policymakers want evidence that survey-based prediction can prioritise limited resources. Patients want earlier risk awareness, used responsibly and with privacy protected.

Overall, stakeholders expect a practical screening tool.

Business Question

The projects business question is can health and lifestyle indicators identify people at higher risk of diabetes so screening and prevention can be targeted more effectively?

The business value is being able to better triage patient with limited screening resources. As discussed above, late diagnosis raises long-term costs, but screening everyone is simply not practical. We attempt to build a model that flags higher-risk people for earlier intervention. We focus on recall and false negatives rather than accuracy alone, while recognising that too many false positives would make the model impractical for deployment.

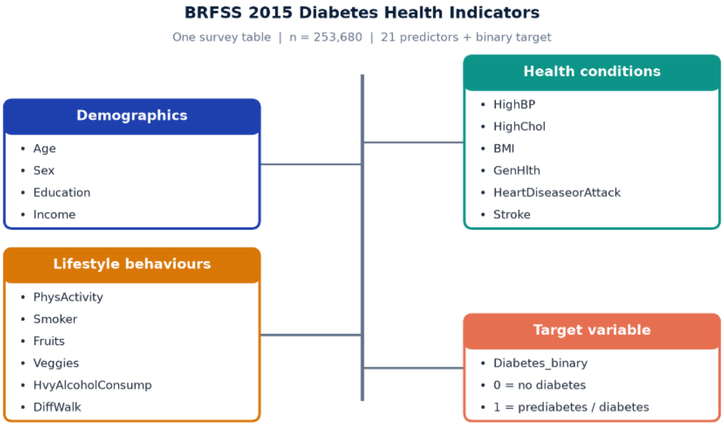
Data Science Question

Data question: to what extent can features such as BMI, blood pressure, cholesterol, general health, physical activity, and demographics predict diabetes or prediabetes?

We use the CDC BRFSS 2015 diabetes health indicators dataset. The target is binary: 0 = no diabetes and 1 = prediabetes or diabetes. The data does not distinguish type 1 from type 2 and combines prediabetes and diabetes in one class.

About The Dataset

The data came from CDC BRFSS 2015 via the UCI Machine Learning Repository (Dataset ID 891). It has 253,680 U.S. survey responses, 21 predictors, and a binary diabetes target (see Figure 3).



The data is useful, but it is not perfect. BRFSS is based on self-reported responses, so it is subject to biases which had to be considered throughout the project.

The quality of the raw data was good for a large survey dataset. There were no missing values, and the variables were well structured for machine learning. The limitations in the data set were that prediabetes and diabetes are combined into one class, with no distinction between type 1 and type 2 diabetes.

The data was generated through the CDC's BRFSS programme, which is a large telephone survey of U.S. adults and is conducted annually, so later surveys could be used for retraining, validation, or updating the model over time.

Data Science Process

Data Analysis

The BRFSS 2015 data was loaded from the UCI Machine Learning Repository (Dataset ID 891) into a Jupyter notebook. The features and target were combined into one pandas DataFrame, column names were standardised, and the data was checked before EDA.

Data wrangling pipeline

1. Load the dataset using fetch_ucirepo(id=891) and merge features and target with pd.concat.
2. Inspect shape, data types, missing values, and unique values against the UCI data dictionary.

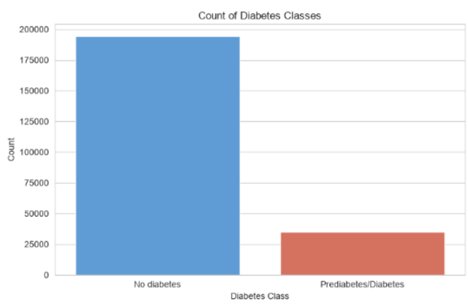
3. Rename columns to snake_case (e.g. HighBP → high_bp, Diabetes_binary → diabetes).
4. Run quality checks: confirm no missing values and check coded variables against expected BRFSS ranges.
5. Remove duplicate rows: 24,206 rows removed (9.54%), leaving 229,474 rows.
6. Group features into binary, ordinal, and numeric types for targeted EDA.
7. Run EDA: distributions, group comparisons, correlation, VIF, and statistical tests.

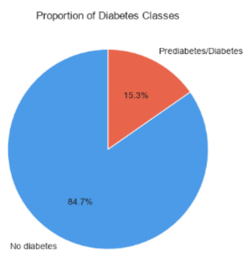
Intermediary data structures
The main working object was a cleaned pandas DataFrame (df). We also used feature-type lists (binary_cols, ordinal_cols, numeric_cols), summary tables (e.g. target_summary), engineered columns (bmi_category, risk_factor_count), groupby/pivot tables, and analysis outputs including the correlation matrix, VIF results, and chi-squared/Cramer's V tables.

EDA Highlights

Class Imbalance

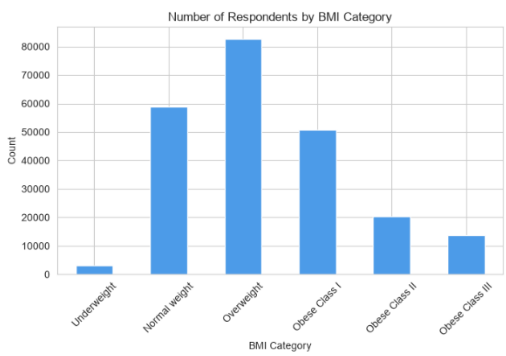
After cleaning the data, the target variable was clearly imbalanced: 84.7% did not have diabetes and 15.3% did (see the below). Modelling therefore had to account for that imbalance, and evaluation could not focus on accuracy alone.

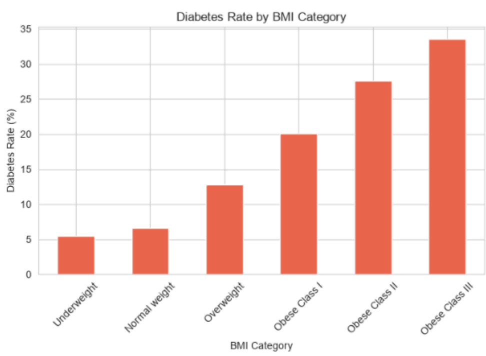




Diabetes Rate By Body Mass Index (BMI)

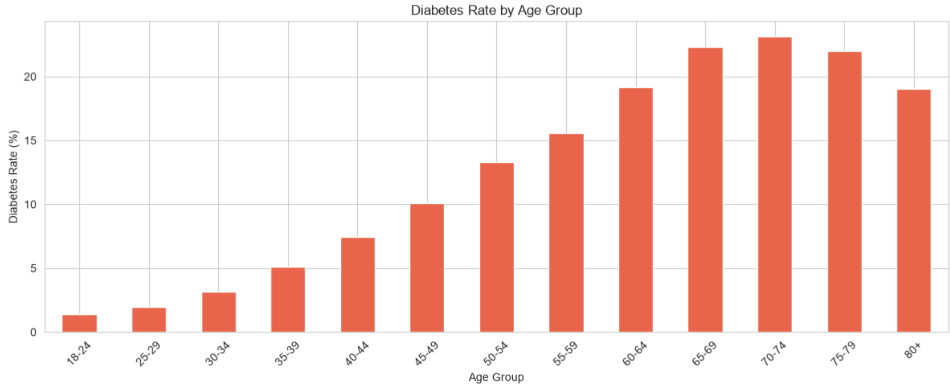
BMI showed a clear pattern. As BMI category increased, diabetes rate increased (see below). BMI was right-skewed, but high values were kept as valid observations**.**

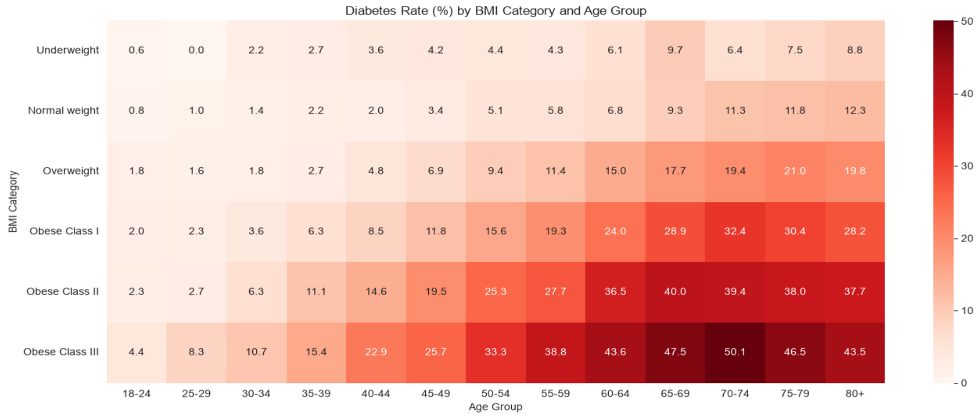




Diabetes Rate By Age Group

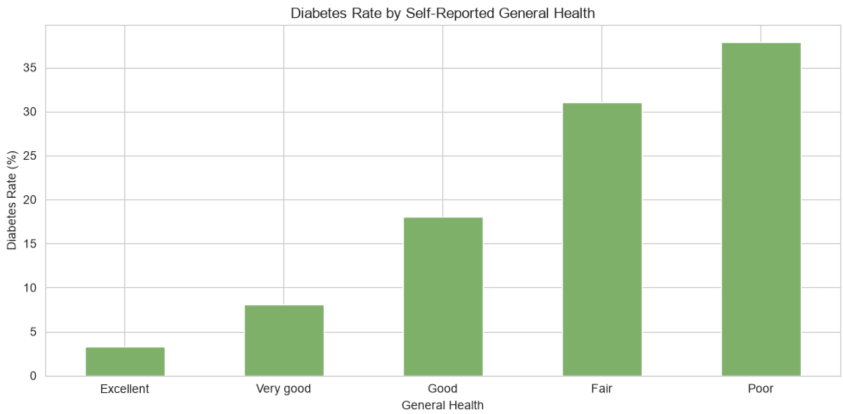
Age followed a similar trend to BMI, the diabetes rate increased with age (see below) Combined with BMI, the highest rates appeared in older respondents with higher BMI (see below).





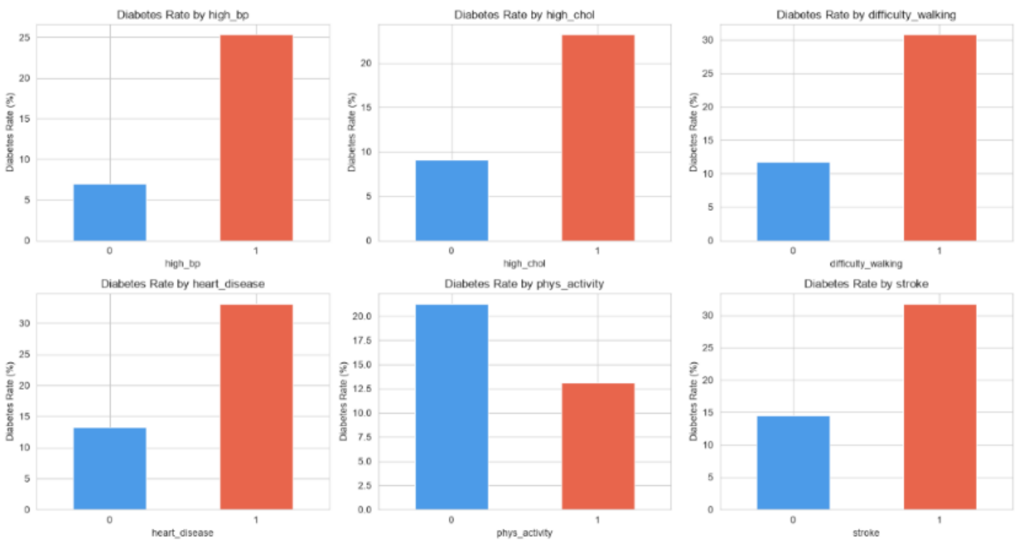
General Self-Reported Health

Respondents reporting poorer health had much higher diabetes rates (see below).



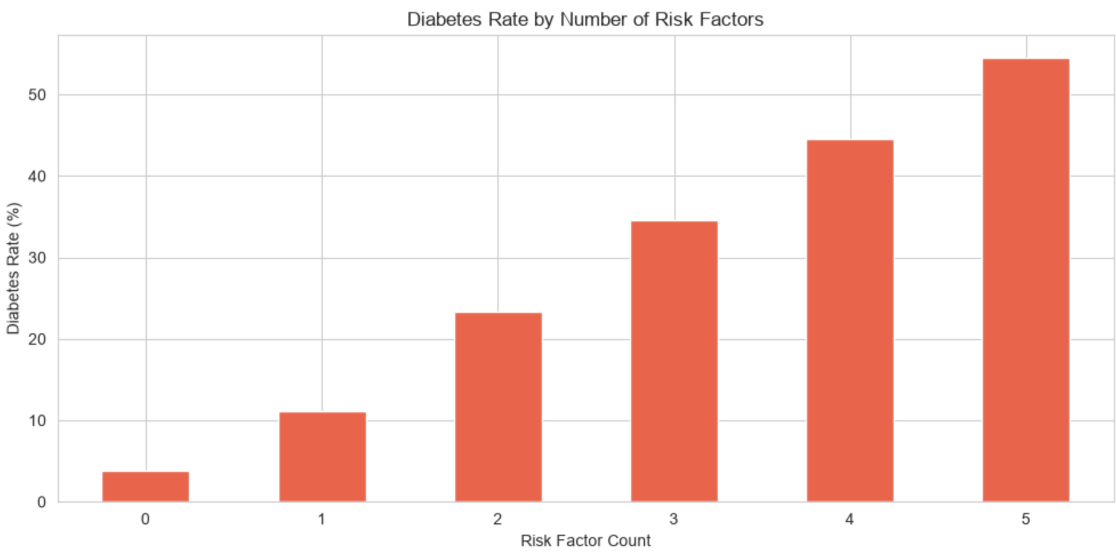
Binary Health Features

Binary health features showed that high blood pressure, high cholesterol, heart disease, stroke, and difficulty walking were more common in the diabetes group. Physical activity was lower in the no diabetes group. (see below). It is also important to note that 0= no diabetic 1= prediabetic/diabetic group.



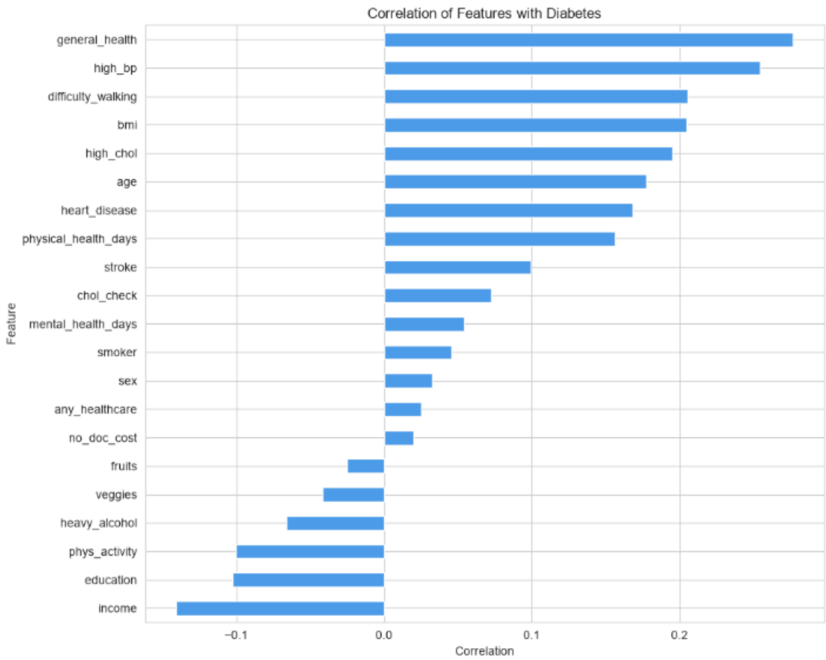
Diabetes By Risk Count

We also created an engineered feature combining the factors in above: high blood pressure, high cholesterol, heart disease, stroke, and difficulty walking. Diabetes rate rose as more risk factors were present (see below) This was treated cautiously in modelling because it partly summarises variables already in the dataset.



Correlation Analysis

Correlation analysis and statistical tests pointed to the same strongest predictors: general health, BMI, age, high blood pressure, and difficulty walking (see below). VIF testing showed limited multicollinearity overall, with some overlap between general health, difficulty walking, and physical health days. Because the dataset is very large, we used Cramer's V and point-biserial correlation rather than p-values alone to rank feature importance.



Conclusion OF EDA Highlights

Overall, the EDA showed that diabetes risk is driven by a mix of physical, lifestyle, and socio-economic factors rather than one variable alone. Modelling therefore needs to consider how features interact, not only how they perform independently.

Pipeline Reusability

The notebook workflow is reusable for future BRFSS data. We would apply the same approach: load the data, rename the variables, validate the groups, and explore the data. Engineered features such as bmi_category and risk_factor_count were exploratory; the original feature sets already formed part of the survey.

Modelling

We modelled diabetes risk as a binary classification problem using three feature sets:

- **Feature Set A** — 21 original cleaned BRFSS variables
- **Feature Set B** — 27 predictors (original variables plus engineered features)
- **Feature Set C** — 14 predictors selected from the EDA (reduced set)

Features in this report simply means predictors (the predictors that influence the target variable).

Feature Set	Predictors	Description	Models Tested
A	21	All original cleaned BRFSS variables	9
B	27	Original variables plus engineered features	9
C	14	Reduced EDA subset with strongest target relationships	9

How We Selected Features

Features were selected from the EDA using diabetes rates by group, correlation with the target, Cramer's V and point-biserial tests, and practical interpretability for screening. Feature Set A was the full set of original features. Feature Set B tested whether engineered features improved model performance. Feature Set C kept only the strongest EDA predictors (14 features).

Feature engineering techniques used:

- Created bmi_category using clinical cut-offs, then one-hot encoded the categories.
- Created risk_factor_count by combining features with the strongest relationships with diabetes.
- Created an overall_health_burden category (general health + mental/physical health days).
- Used StandardScaler for models that need scaled inputs.
- Used class_weight='balanced' / imbalanced-learn methods to account for class imbalance.
- Removed raw bmi from Feature Set B to avoid duplication when testing the models.

The Models Used

Nine models were trained on each feature set (27 models in total). We started with standard classifiers such as Logistic Regression, Linear SVM, RBF SVM, and Gaussian Naive Bayes. We then used ensemble methods and approaches designed for class imbalance, including Balanced Bagging, Balanced Random Forest, and Easy Ensemble. Finally, we tested Voting and Stacking classifiers to see whether combining models improved performance.

Training time

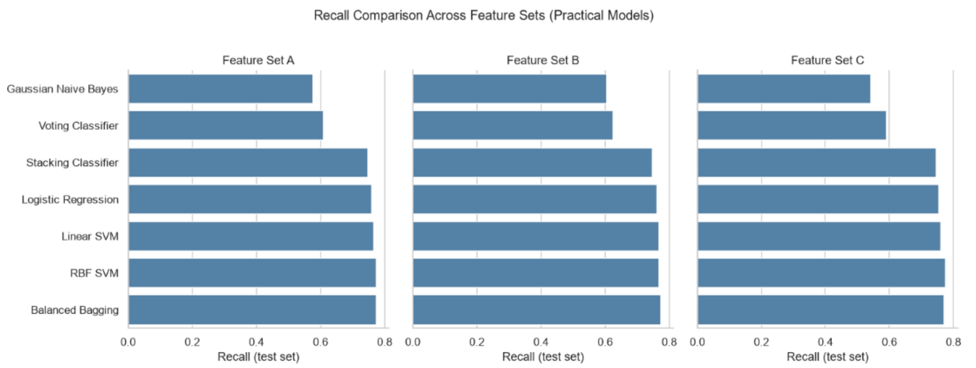
Linear models were trained on the full training set and were relatively fast. Slower models (RBF SVM, Balanced Random Forest, Easy Ensemble) used a stratified 20,000-row training subsample. Voting and Stacking used a 30,000-row subsample. Balanced Bagging used the full training set. This kept runtime practical on a local machine while still allowing fair comparison across models.

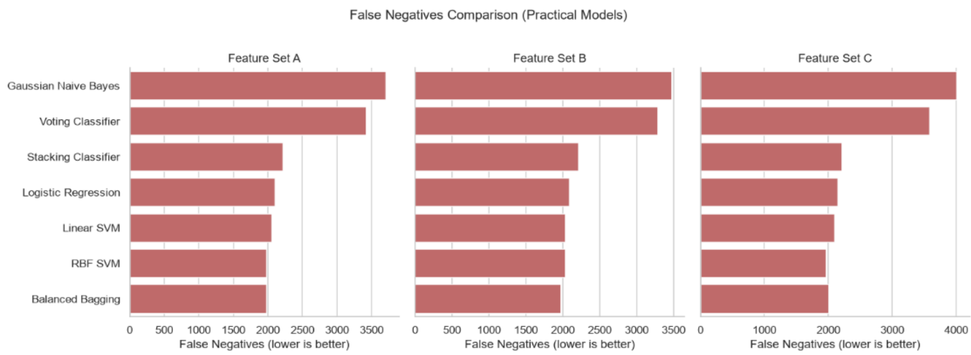
Tools used

Modelling was done locally in Python / Jupyter notebooks, using pandas, scikit-learn, imbalanced-learn, matplotlib, and seaborn. No cloud platform was required.

Model performance metrics

Because the target variable is imbalanced, we evaluated recall, false negatives, precision, false positives, accuracy, F1-score, and ROC AUC. Recall and false negatives were the priority metrics for evaluating the models (See below).





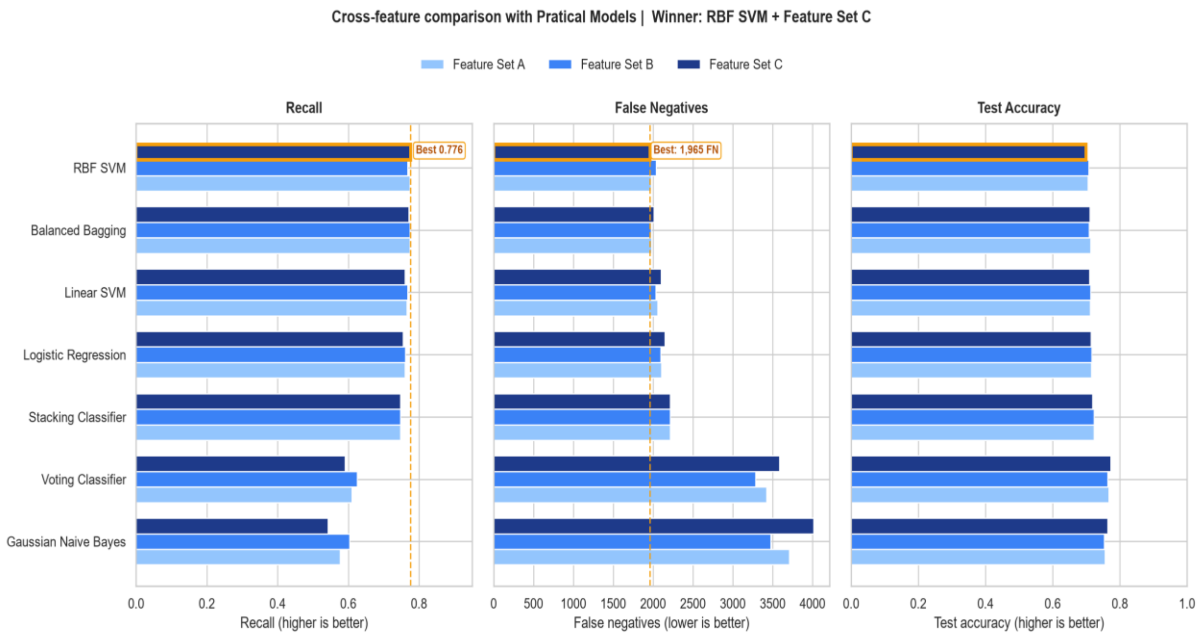
Feature Set That Captured Most Of The Performance

Feature Set C (14 predictors) performed similarly to Sets A and B, and better on some models. RBF SVM on Feature Set C had the best recall in the project (0.776, with 1,965 false negatives). The Figure below shows the best model for each feature set. Performance was close across all three features, but RBF SVM on Feature Set C was slightly better on recall and false negatives, so that was the model we selected.

Feature Set	Model	Test Acc.	Recall	F1	ROC AUC	FP	FN
A	RBF SVM	0.7048	0.7746	0.4453	0.8034	14,955	1,978
B	Balanced Bagging	0.7083	0.7746	0.4482	0.8107	14,758	1,978
C	RBF SVM	0.6994	**0.7760**	0.4412	0.7984	15,281	**1,965**

Selected Model

After comparing each model across the feature sets, we concluded that Feature Set C with RBF SVM gave the best recall and the fewest false negatives. It is worth noting that accuracy was only 0.6994. Other models were more accurate but had lower recall and more false negatives, which we were trying to avoid. Most models did not perform as well as we hoped on accuracy and precision. If we were to do the project again, we would spend more time on hyperparameter tuning and testing additional models to improve accuracy and increase recall further.



Modelling Conclusion

There was a practical trade-off in modelling, higher accuracy usually meant lower recall, and higher recall usually meant lower accuracy. None of the models performed exceptionally well, but RBF SVM on Feature Set C was the most balanced overall. Some ensemble models improved recall without improving accuracy, and Feature Set B's engineered features did not meaningfully lift performance.

Outcome

The main finding from completing the data science process is that health and lifestyle survey data can support diabetes risk screening and that there are multiple features that predict diabetes.

Feature Set C showed that 14 survey predictors could capture most of the predictive performance without needing all the original features or the engineered features in Feature Set B.

Overall, the data science process suggests that deploying a smaller Feature Set C model could be useful for screening people likely to be at risk of diabetes. The conclusion is not that diabetes can be predicted with high accuracy from survey data alone. It is that survey indicators can flag higher-risk people earlier, provided the model is judged on recall and missed cases rather than accuracy alone.

Implementation

BRFSS is updated annually, so the model would need a repeatable pipeline to load new survey data, apply the same cleaning and Feature Set C transformations, and score respondents accordingly. The output should be a screening risk flag for follow-up testing, not a diabetes diagnosis. That means clinicians or public health teams could review flagged cases before deciding whether a patient needs further testing to confirm diabetes.

Another consideration is storing and accessing the data. Survey extracts, scored outputs, and model versions would need to be stored securely, with clear audit trails for who was flagged and why. Because the selected model is RBF SVM on 14 predictors, the production feature set should stay limited to those variables, so the deployed model remains consistent with what was tested.

Finally, the model would need ongoing monitoring and further hyperparameter tuning so that accuracy, precision, recall, and F1-score can improve. If performance dropped with future survey data, the model would need to be retrained, or its parameters retuned before wider deployment. A good first step would be a pilot with a GP clinic or public health organisation rather than immediate full-scale deployment.

Data Answer

The data science question was answered satisfactorily for a screening use case, but not for high-accuracy prediction.

The models were not as accurate as we had hoped. Precision was also limited, so many people flagged as higher risk would still need confirmatory testing. However, across models, BMI, age, and general health were among the strongest predictors, along with high blood pressure and high cholesterol.

Our confidence in the data answer is therefore moderate. We are confident that health and lifestyle survey features can support screening triage, and that the key predictors are stable across models and feature sets. We are, however, less confident that the current models are strong enough for high-precision or diagnostic use without further hyperparameter tuning, validation on later BRFSS years, and clinical review of flagged cases.

Business Answer

The business question was answered satisfactorily for screening triage, but not as a ready-to-deploy diagnostic solution.

From a business point of view, the results of the project are useful, but the model needs retuning before deployment. Therefore, our confidence in the business answer is similar to the data answer, being that we are less confident that the model is ready for full-scale rollout without further tuning, validation on later BRFSS years, and a pilot with a GP clinic or public health organisation.

Response To Stakeholders

The overall response to the project's stakeholders is that the project is promising for diabetes screening triage, but it is not ready to replace clinical diagnosis.

The next step should be a pilot survey with a GP clinic or public health organisation, using the model as a risk flag for follow-up testing rather than as a diagnostic decision. Stakeholders should treat BMI, age, general health, high blood pressure, and high cholesterol as the strongest practical indicators to monitor. Before wider public use, the model should be validated on later BRFSS years and hyperparameter tuned further to improve accuracy and precision while protecting recall.

End To End Solution

The end-to-end solution was to create a screening pipeline, not a diagnostic system.

Before deployment, new BRFSS survey data would be acquired and stored securely. The same cleaning steps used in this project would be applied, and only the Feature Set C predictors would be used. Those features would be passed into the trained RBF SVM model, which would output a diabetes risk flag for each respondent.

Flagged cases would then be reviewed by clinicians or public health staff before any further action. Where appropriate, people would then be referred for confirmatory testing such as HbA1c or fasting glucose, then for prevention support or treatment depending on the result. Model performance would be monitored over time, and the pipeline would be rerun when later BRFSS survey data become available.

References For Data

Item	Location / description
Dataset	CDC Diabetes Health Indicators (UCI ID 891), based on BRFSS 2015
Dataset link	https://archive.ics.uci.edu/dataset/891/cdc+diabetes+health+indicators
EDA notebook	CDM_EDA_Capstone_FINAL.ipynb
Feature engineering notebook	Final Feature Selection and Engineering.ipynb
Modelling notebooks	Feature A Modelling.ipynb; Feature B Modeling.ipynb; Feature C Modelling.ipynb
Comparison notebook	Feature Sets A B C Comparison and Conclusions.ipynb
Exported results	all_model_results_a/b/c.csv and all_model_results_combined.csv
Exported models	No separate model files; models trained in notebooks, metrics saved to CSV
Project folder	c:\Users\Conno\Data Science Course\Capstone\\

The Figure above shows where the notebooks are stored and the notebooks that went into this project.

Resources used in the project

Libraries

- Python / Jupyter Notebooks
- pandas, NumPy
- matplotlib, seaborn
- scikit-learn
- imbalanced-learn
- ucimlrepo
- SciPy

Algorithms / models

- Logistic Regression
- Linear SVM
- RBF SVM
- Gaussian Naive Bayes
- Balanced Bagging

- Balanced Random Forest
- Easy Ensemble
- Voting Classifier
- Stacking Classifier

Other methods

- Train/test split with stratification
- StandardScaler
- One-hot encoding
- Class weighting / imbalance-aware learning
- Evaluation metrics: accuracy, precision, recall, F1-score, ROC AUC, false positives, and false negatives

References For Journal Articles

Bajaj, M., McCoy, R. G., Balapattabi, K., Bannuru, R. R., Bellini, N. J., Bennett, A. K., Beverly, E. A., Briggs Early, K., Brown, F. M., Callaghan, B. C., ChallaSivaKanaka, S., Das, S. R., Dixon, D. L., Drincic, A., Ebekozien, O., Echouffo-Tcheugui, J. B., Everett, B. M., Frykberg, R. G., Garg, R., … ElSayed, N. A. (2025). Summary of Revisions: Standards of Care in Diabetes—2026. *Diabetes Care*, *49*(Supplement_1), S6–S12. https://doi.org/10.2337/dc26-srev

Centers for Disease Control and Prevention. (n.d.). *Behavioral Risk Factor Surveillance System*. https://www.cdc.gov/brfss/index.html

Chatterjee, R., Narayan, K. M. V., Lipscomb, J., & Phillips, L. S. (2010). Screening adults for pre-diabetes and diabetes may be cost-saving. *Diabetes Care, 33*(8), 1694–1700. https://doi.org/10.2337/dc10-0054

Corkey, B. E. (2012). Diabetes: have we got it all wrong? Insulin hypersecretion and food additives: cause of obesity and diabetes?. *Diabetes care*, *35*(12), 2432.

Inzucchi, S. E. (2012). Diagnosis of diabetes. *New England journal of medicine*, *367*(6), 542-550.

Majcherek, D., Ciesielski, A., & Sobczak, P. (2025). AI-driven analysis of diabetes risk determinants in U.S. adults: Exploring disease prevalence and health factors. *PLOS ONE, 20*(9), Article e0328655. https://doi.org/10.1371/journal.pone.0328655

Papatheodorou, K., Banach, M., Bekiari, E., Rizzo, M., & Edmonds, M. (2018). Complications of diabetes 2017. *Journal of Diabetes Research*, *2018*(3086167), 1–4. https://doi.org/10.1155/2018/3086167

Xie, Z., Nikolayeva, O., Luo, J., & Li, D. (2019). Building risk prediction models for type 2 diabetes using machine learning techniques. *Preventing Chronic Disease, 16*, Article 190109. https://doi.org/10.5888/pcd16.190109